
3AL Tutorial 9 2024

1. (a) Let A and B be distinct subgroups of a group G . Show that if $|A| = |B| = p$ then $|AB| = p^2$, that is, every element of the form ab (where $a \in A$ and $b \in B$) is distinct.

HINT: Suppose there are some $a_1b_1, a_2b_2 \in AB$ such that $a_1b_1 = a_2b_2$. Rearrange this to find an element $x \in A \cap B$. Show that x must be the identity and so $a_1 = a_2, b_1 = b_2$.

- (b) Let G be a group of order pq for some primes p and q with $p > q$. Let $a \in G$ be an element of order p .

(i) Why does such an element a exist?

(ii) Use (a) to show that the only subgroup of G of order p is $A = \langle a \rangle$.

(iii) Prove that $A \triangleleft G$.

- (c) Prove or disprove: There is a simple group of order 15.

NOTE: Recall that a group is called simple if its only normal subgroups are itself and its trivial subgroup.

2. Let G be a p -group. Use the Class Equation to prove that $Z(G)$ is nontrivial.
3. Let G be a non-abelian p -group. Show that G has an inner automorphism of order p .

HINT: The order of an inner automorphism is just the order of the inner automorphism when considered as an element of $\text{Inn}(G)$. Recall that $\text{Inn}(G) \cong G/Z(G)$.

4. Let G be a p -group. Suppose G acts on a set X , show that p divides $|X| - |X_f|$.
5. Let G be a p -group and H a nontrivial normal subgroup of G . Show that $H \cap Z(G)$ is nontrivial.

HINT: Consider the group action of conjugation by G on H .

6. Let G be a non-abelian group of order p^3 for some prime p . Prove that $|Z(G)| = p$ and $G/Z(G) \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

HINT: Recall that if $G/Z(G)$ is cyclic, then G is abelian.

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7. Let G be a non-abelian p -group. Show that p^2 divides $|G : Z(G)|$.
8. (a) Let G be a p -group. Show that the number of subgroups of order p^k is congruent modulo p to the number of normal subgroups of order p^k .
- (b) Let G be a p -group. Show that the number of non-normal subgroups of G is a multiple of p .

HINT: Define some group actions and use the fact that $|X| \equiv |X_f| \pmod{p}$.

9. Let G be a group with normal subgroup K . Prove that if G/K is a p -group, then G has a normal subgroup of index p .
10. What are the orders of the Sylow p -subgroups of a group of order 180? What about a group of order 700?
11. Find all Sylow p -subgroups of S_3 for $p = 2$ and $p = 3$. Confirm the following (for $p = 2$ and $p = 3$, separately):
- (a) Any two Sylow p -subgroups are conjugate.
- (b) The number of Sylow p -subgroups is congruent to 1 modulo p .
- (c) The number of Sylow p -subgroups divides $|S_3|$.
12. Let G be a group with order $p^n m$ where p is a prime with $p \nmid m$. Suppose H is a Sylow p -subgroup of G . Prove or disprove: Any conjugate of H is a Sylow p -subgroup.